Can a Portfolio Be Held to a Tail-Loss Budget?

Minimum-CVaR Rules, Expected-Shortfall Forecasts
and a Test on Data They Had Not Seen

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Abstract. In this study, I ask whether a portfolio can be held to a fixed limit on its expected loss on its worst days. In a first version (Part I), I compared five portfolio rules, including minimum Conditional Value-at-Risk (CVaR), on twelve assets measured in Indian rupees. Equal weighting had the highest Sharpe ratio, and the minimum-CVaR rule's best result, a gain in 2008, rested on weights fitted on later data. In Part II, I treat CVaR as a forecast and test it. On eight rupee asset classes from 2006 to 2026, forecasts that rescale past returns by current volatility score significantly better than plain historical simulation, and their breaches cluster far less, though a strict joint test rejects every model. A portfolio optimised each week under a budget of 1.5% for its expected shortfall at 97.5% realised 1.51% in the development period and 1.69% in a holdout run once, an excess that is not significant. It lost −2.2% in the 2008 crisis against −22.9% for a 60/40 portfolio, and its Sharpe ratio was higher than 60/40's but not significantly; the sample could only have detected a gap three times as large. I then ran this method, unchanged, on data it had not seen: nine series back to 1928 and a dollar investor from 1964. The forecasts' advantage replicated. The portfolio again held its budget, but it matched rather than beat 60/40 and lost more in the slow bear market of 1973-74.

1. Introduction

An investor who says "I can live with losing 1.5% on a bad day, but not much more" is stating a limit on tail risk, not on volatility. The natural measure of that limit is Conditional Value-at-Risk (CVaR), also called expected shortfall (ES): the average loss on the worst days. Since Rockafellar and Uryasev (2000) showed that minimising CVaR over a sample of returns is a linear program, it has been a practical portfolio objective. Whether it works out of sample is a separate question, and the evidence is not encouraging: Lim et al. (2011) show that the minimum-CVaR solution is fragile under estimation error, and DeMiguel et al. (2009) found that no sample-based optimised portfolio they examined consistently beat equal weighting.

My first version of this study (Part I) compared five portfolio rules, including minimum CVaR, on twelve assets measured in rupees from 2015 to 2026. Equal weighting had the highest Sharpe ratio out of sample; minimum CVaR had the smallest drawdown and a gain in a replay of the 2008 crisis. When I reread it, I found that the design could not support its strongest claims. The 2008 replay used weights estimated on 2015–2026 data, so it was not out of sample. The safe asset was a US bond fund, which for a rupee investor is mostly a position in the exchange rate. Most importantly, the study never asked whether its CVaR estimates were right: CVaR is a forecast of tail loss, and it was never tested as one.

*Email: arjunprashanth6129@gmail.com. Code, data and the scripts that generate every number, table and figure in this paper are at https://github.com/arjunprashanth6129/cvar-portfolio-optimizer. An interactive summary of the results is at https://arjun-quant-projects.vercel.app/cvar.

In Part II, I rebuild the study around that question. Part II-A tests five forecasts of tomorrow's Value-at-Risk (VaR) and ES, from plain historical simulation to models that rescale history by current volatility, with formal backtests. Part II-B uses the forecast to run a portfolio to a fixed budget: each week, the portfolio may hold whatever mix of eight rupee asset classes maximises a simple measure of expected return, provided its forecast ES for the next day does not exceed 1.5%, with the rest in a liquid fund. Every choice was written down and committed to version control before any portfolio return was computed, the one choice made on data was made on a development period ending in 2016, and the holdout from 2017 was run once.

After the single run I added two things, also written down before they were computed. An audit applied stricter tests to the results, and a replication ran the method, unchanged, on data it had never seen: nine long series back to 1928, and a dollar investor from 1964 to 2026.

The results are mixed, and I report them as they came out. The forecasting result is strong: filtered forecasts beat historical simulation in every sample, including a century of unseen data, and their breaches cluster less than half as much, although a strict joint test rejects every model. The portfolio held its budget in both markets and lost far less than 60/40 in crashes where volatility rose quickly, including 2008. It did not reliably beat 60/40 on return per unit of risk: in India the sample was too short to tell, and for the dollar investor over 64 years the two were equal, with the budget portfolio losing more in the slow decline of 1973-74.

Section 2 summarises Part I and what it could not show. Section 3 sets out the design of Part II, Sections 4 and 5 report its results and Section 6 its robustness. Section 7 reports the audit and Section 8 the replication. Sections 9 and 10 discuss the findings and their limits, and Section 11 concludes.

2. Part I: five rules on twelve assets

2.1 Value-at-Risk, CVaR and the linear program

Let w be a vector of portfolio weights, r a vector of daily asset returns and $L = -w^\top r$ the one-day loss. At confidence level α, Value-at-Risk is the α-quantile of the loss,

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : \Pr(L \le \ell) \ge \alpha\}, \tag{1}$$

and CVaR, or expected shortfall, is the expected loss beyond it,

$$\mathrm{ES}_\alpha(L) = \mathbb{E}\left[L \mid L \ge \mathrm{VaR}_\alpha(L)\right]. \tag{2}$$

ES is a coherent risk measure (Artzner et al., 1999; Acerbi and Tasche, 2002), so diversification never increases it. For N equally likely scenarios with losses $L_j(w) = -r_j^\top w$, Rockafellar and Uryasev (2000, 2002) show that

$$\mathrm{ES}_\alpha(w) = \min_\zeta \; \zeta + \frac{1}{(1-\alpha)N} \sum_{j=1}^{N} \left(L_j(w) - \zeta\right)^+, \tag{3}$$

which, with an auxiliary variable u_j for each positive part, is the linear program

$$\min_{w,\,\zeta,\,u} \; \zeta + \frac{1}{(1-\alpha)N} \sum_{j=1}^{N} u_j \quad \text{s.t.} \quad u_j \ge -r_j^\top w - \zeta, \;\; u_j \ge 0, \;\; \mathbf{1}^\top w = 1, \;\; w \ge 0. \tag{4}$$

I solve it with CVXPY (Diamond and Boyd, 2016) and the Clarabel solver (Goulart and Chen, 2024). Part II uses the same program with the ES as a constraint rather than the objective.

Table 1. The five portfolio rules of Part I

Rule	Objective	Inputs estimated
Equal weight	$w_i = 1/N$	none
Minimum variance	minimise $w^\top \Sigma w$	covariance matrix
Maximum Sharpe	maximise $(\mu^\top w - r_f)/\sqrt{w^\top \Sigma w}$	covariance and expected returns
Risk parity	equalise $w_i(\Sigma w)_i$ across assets	covariance matrix
Minimum CVaR	minimise $\mathrm{ES}_{0.95}$ of daily loss, eq. (4)	historical return sample

All portfolios are long-only and fully invested. Maximum Sharpe is solved by the transformation of Charnes and Cooper (1962) and risk parity in the convex form of Spinu (2013).

Table 2. Part I: in-sample and out-of-sample performance of the five rules

	In sample, full period				Out of sample, walk-forward, net of costs					
Rule	Return	Vol.	CVaR	Sharpe	Return	Vol.	CVaR	Sharpe	Max DD	Turnover
Equal weight	19.03	11.68	1.625	1.115	21.21	12.21	1.710	**1.245**	21.72	4.7
Risk parity	14.26	8.70	1.189	0.949	15.68	9.01	1.243	1.075	17.28	4.0
Maximum Sharpe	24.83	13.56	1.895	1.388	19.33	16.04	2.389	0.832	34.36	25.4
Minimum CVaR	8.48	6.25	0.836	0.397	11.05	6.72	0.913	0.751	**12.19**	7.2
Minimum variance	8.73	6.22	0.842	0.439	10.66	6.55	0.882	0.712	12.72	3.9
Unoptimised references										
Nifty ETF, buy and hold					13.00	15.03	2.232	0.466	36.34	
Equal weight, never rebalanced					30.61	37.72	5.552	0.652	57.61	

Returns, volatility, maximum drawdown and turnover are in percent; CVaR is the daily historical CVaR at 95%, in percent. In-sample statistics use 5 January 2015 to 14 August 2026, with arithmetic mean returns. Out-of-sample statistics use the walk-forward test from 31 March 2017 to 14 August 2026, with compound returns net of 20 basis points per unit traded. The Sharpe ratio uses a risk-free rate of 6%.

2.2 Design and results

Part I compared the five rules in Table 1 on twelve assets: an Indian equity ETF, three Indian stocks, gold in rupees and in dollars, four US-listed equity and bond ETFs, and Bitcoin, from 5 January 2015 to 14 August 2026. Dollar prices were converted to rupees at the daily exchange rate. Each rule was re-estimated at every quarter-end on the trailing 500 trading days and charged 20 basis points per unit traded, giving 38 out-of-sample rebalances from 31 March 2017 to 14 August 2026.

Table 2 and Figure 1 report the results. Out of sample, equal weighting had the highest Sharpe ratio, 1.245, and minimum CVaR the smallest maximum drawdown, 12.19%. Maximum Sharpe, the only rule that estimates expected returns, was the only one whose Sharpe ratio fell out of sample, from 1.388 to 0.832, the error maximisation that Michaud (1989) describes. None of equal weighting's Sharpe-ratio differences from the optimised rules was significant (Jobson-Korkie p between 0.086 and 0.272). Applied to the returns of the 2008 crisis, the minimum-CVaR weights gained +5.13% while equal weighting lost 18.45%.

2.3 What Part I could not show

Four features of that design limit what it can support.

- *The 2008 result was not out of sample.* The crisis replay applied weights estimated on 2015–2026 to returns from 2008, so the portfolio was built with knowledge of the decade that followed.

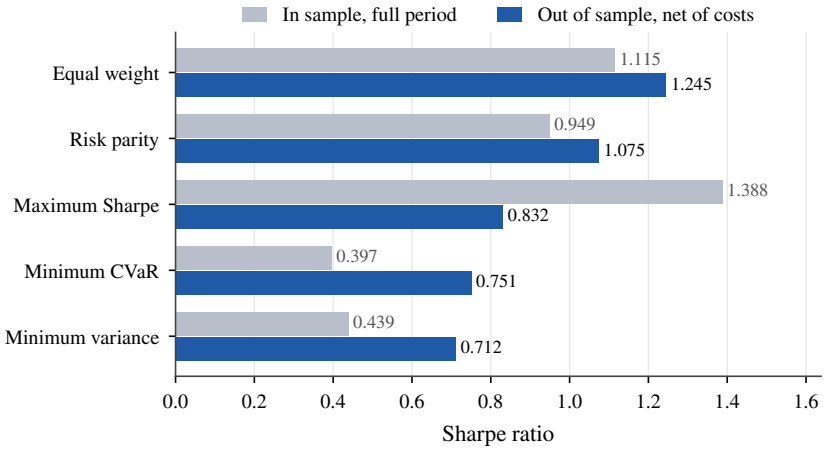


Figure 1. Part I: Sharpe ratio of each rule in sample and out of sample, net of costs.

- *The safe asset was the exchange rate.* The universe had no Indian bonds or cash, so the minimum-risk rules held most of their weight in a US bond fund, 67% of it for minimum CVaR. Most of their 2008 gain came from the rupee's fall of 16.7% against the dollar, which raised the rupee value of that fund.
- *The risk estimate was never tested.* A CVaR estimate is a forecast of tomorrow's tail loss, and Part I never checked whether the forecasts were right.
- *The comparison mixed risk levels.* The rules' volatilities out of sample ranged from 6.5% to 16.0% a year, so ranking them by Sharpe ratio answered a different question from the one CVaR is built for.

Part II is designed around these four points.

3. Part II: design

3.1 Universe and data

The universe has eight risky asset classes and cash, chosen by asset class rather than by performance (Table 3). Indian equity is the Nifty 50, Nifty Midcap 150 and Nifty Smallcap 250 total return indices, less an assumed index-fund fee of 0.10% a year for the Nifty 50 and 0.30% for the other two. Indian bonds are a 10-year constant-maturity gilt fund and cash is a liquid fund, both from the net asset values (NAVs) published by the Association of Mutual Funds in India, so they are net of the funds' expenses. The liquid fund is also the risk-free rate. Gold is Gold BeES, the oldest Indian gold ETF, chained across its three fund houses; before its first NAV in March 2007 (239 sessions) it is the US gold ETF GLD converted to rupees. Foreign assets, which a resident can buy under the Liberalised Remittance Scheme, are the S&P 500 (SPY), developed markets outside North America (EFA) and US Treasuries of 7–10 years (IEF). The sample runs from 3 April 2006 to 30 September 2026, 5,082 sessions of the National Stock Exchange (NSE).

Timing. A decision is made on the evening of session t, after the funds publish that day's NAVs, and filled at session $t + 1$'s prices. A US price enters session t only if it was stamped before t's calendar date: the US close of $t - 1$ happens at 01:30 India time on day t, before the Indian close, while the US close of t does not. Dollar prices are converted at the Federal Reserve's H.10 noon rate of the previous business day.

Data checks. Before any forecast was computed, I checked every series against an independent source. The first exchange rate I used, Yahoo Finance's, was up to 6.3% away from both the Federal Reserve's and the

Table 3. Part II: the eight asset classes and cash, 3 April 2006 to 30 September 2026

Asset class	Source	Return (%)	Volatility (%)	Daily ES 97.5% (%)	Worst day (%)
Nifty 50	TRI	11.0	20.9	4.03	-12.9
Nifty Midcap 150	TRI	14.2	21.0	4.27	-13.0
Nifty Smallcap 250	TRI	12.9	21.6	4.46	-12.4
10-year gilt fund	AMFI NAV	7.9	2.8	0.55	-2.9
Gold	AMFI NAV	14.0	17.4	3.24	-10.0
S&P 500	SPY	15.6	19.0	3.63	-10.8
Developed ex-US	EFA	9.6	20.6	4.02	-11.1
US Treasuries 7–10y	IEF	7.2	10.3	1.71	-4.6
Liquid fund (cash)	AMFI NAV	7.1	0.3	0.00	-0.2

Returns are compound annual returns in rupees, volatility is annualised, and ES is the average of the worst 2.5% of daily returns over the whole sample, as a loss. TRI is the NSE Indices total return index; AMFI NAV is the fund's published net asset value; SPY, EFA and IEF are US ETFs converted to rupees. These statistics describe the data and were not used to choose anything.

European Central Bank's rates on some days, enough to create false 6% days in the US Treasury series; the Federal Reserve's and the ECB's rates agree within 0.5% on 98.3% of days, so I switched to the Federal Reserve's. Two funds changed the size of their units, Gold BeES 1:100 on 23 December 2019 and the liquid fund 1:10 on 20 June 2026; both were detected as one-day moves of exactly those ratios and undone. Gold BeES and GLD in rupees have a weekly correlation of 0.925 where both exist; their annual returns differ when India changes its import duty on gold, by −9.8% in 2024. The Nifty 50 total return index exceeds the price index by 1.2% a year, the dividend yield.

3.2 Five forecasts of tomorrow's tail loss

Each model forecasts VaR and ES at $\alpha = 0.975$, the level of the Basel Committee's market-risk standard (Basel Committee on Banking Supervision, 2019), for the next session, using returns up to the current one. From a sample of N returns sorted upwards, with $k = (1-\alpha)N$, VaR is minus the $\lceil k \rceil$-th smallest return and ES is minus the average of the worst k, counting the last at its fractional weight (Acerbi and Tasche, 2002). This is the quantity the program in equation (4) works with, so a forecast and a budget mean the same thing.

- *Historical simulation (HS)*: the empirical VaR and ES of the last $N = 500$ returns, Part I's estimator.
- *Normal, EWMA*: a normal distribution with zero mean and the exponentially weighted volatility of RiskMetrics (J.P. Morgan and Reuters, 1996), $\sigma^2_{t+1} = \lambda\sigma^2_t + (1-\lambda)r^2_t$ with $\lambda = 0.94$.
- *Filtered historical simulation* (Hull and White, 1998; Barone-Adesi et al., 1999): each past return is divided by the volatility forecast for its own day, $z_s = r_s/\sigma_s$, and the last 500 of these standardised returns are multiplied by tomorrow's volatility, $\sigma_{t+1}z_s$. The tail keeps its historical shape but moves with the current level of risk. I use two volatility models: EWMA, and GARCH(1,1) (Bollerslev, 1986),

$$\sigma^2_{t+1} = \omega + a\, r^2_t + b\, \sigma^2_t, \tag{5}$$

  estimated by Gaussian quasi-maximum likelihood on the last 1,000 returns and refitted at each month-end.

- *GARCH-EVT* (McNeil and Frey, 2000): GARCH standardised losses above their 90th percentile u are fitted with a generalised Pareto distribution with shape ξ and scale β, and

$$\mathrm{VaR}_z = u + \frac{\beta}{\xi}\left[\left(\frac{1-\alpha}{0.10}\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_z = \frac{\mathrm{VaR}_z}{1-\xi} + \frac{\beta - \xi u}{1-\xi}, \tag{6}$$

  each multiplied by σ_{t+1}.

The models are applied to ten test portfolios: each of the eight assets, an equal mix of them, and 60/40 (60% Nifty 50 and 40% gilt).

3.3 Testing a forecast

If a VaR forecast is right, a *hit*, a day whose loss exceeds the forecast VaR, occurs on a share $1-\alpha$ of days, independently of earlier hits. Kupiec's test (Kupiec, 1995) checks the share, and Christoffersen's conditional-coverage test (Christoffersen, 1998) checks the share and the independence together. Acerbi and Szekely's Z_2 (Acerbi and Szekely, 2014) checks whether losses beyond VaR were as large as the ES forecast said:

$$Z_2 = \frac{1}{T(1-\alpha)} \sum_{t=1}^{T} \frac{r_t \, \mathbf{1}\{r_t < -\mathrm{VaR}_t\}}{\mathrm{ES}_t} + 1, \tag{7}$$

which is zero when the forecasts are right and negative when risk was understated; its p-value is simulated from each day's forecast distribution. To rank models I use the FZ0 loss of Patton et al. (2019), a score for VaR and ES together that the true forecasts minimise (Fissler and Ziegel, 2016), and test the difference in average loss between two models with the test of Diebold and Mariano (1995), using Newey-West standard errors (Newey and West, 1987).

3.4 Running a portfolio to a budget

The mandate is that the forecast ES at 97.5% of tomorrow's return must not exceed $B = 1.5\%$. A normal distribution with 10% annual volatility has a daily ES of about 1.47%, so B is a conventional balanced mandate, fixed without looking at the data. The primary rule, P1, solves each week

$$\max_{w} \sum_{i} w_i \bar{\sigma}_i \quad \text{s.t.} \quad \mathrm{ES}_{0.975}(w) \le B, \;\; 0 \le w_i \le 0.4, \;\; \mathbf{1}^{\top} w \le 1, \tag{8}$$

with the remaining $1 - \mathbf{1}^{\top} w$ in the liquid fund. The ES is computed on *conditional scenarios*: each asset's last 500 standardised returns, from its own GARCH model, multiplied by its volatility forecast for tomorrow, with the days kept aligned so that the dependence between assets is the historical one. The constraint is written in the form of equation (4), so the problem is a linear program. The objective weights each asset by its trailing volatility $\bar{\sigma}_i$. This maximises expected excess return if every asset class earns the same return per unit of risk, so no expected return is estimated, which is the weakness Part I exposed in maximum Sharpe.

A linear program's solution sits at a vertex of its feasible set, and a small change in the scenarios can move it to another vertex. On simulated data the plain solution turned its portfolio over about ten times a year. P1 therefore has a second stage: among the portfolios within 5% of the best objective and inside the same constraints, it takes the one that is cheapest to reach from the current holdings.

The comparison rules run on the same schedule. P2 is P1 on raw historical scenarios, which isolates the effect of filtering. P3 and P4 are equal weight and risk parity, each scaled by $\min(1, B/\widehat{\mathrm{ES}})$. P5 is Part I's minimum-CVaR rule on conditional scenarios, fully invested. The benchmarks are a static 60/40 and a static equal weight across the eight assets, rebalanced quarterly, and matched-exposure versions of P3 and P4, the

unmanaged mix held at the managed version's average exposure, which follows the comparison of Moreira and Muir (2017). The matched versions use the whole period's average exposure and so are not tradable.

Decisions are taken at the last session of each week and filled at the next. A trade is skipped if its one-way turnover would be under 5%, unless the drifted portfolio's forecast ES exceeds the budget. Each side of a trade pays 10 basis points for Indian equity and gold ETFs, 1 basis point for debt and liquid funds, and 50 basis points for foreign assets, which a retail investor buys through a bank remittance.

3.5 Protocol

The development period runs from the first forecast on 1 July 2008 to 30 December 2016 (2,101 sessions) and the holdout from 2 January 2017 to 30 September 2026 (2,416 sessions). The specification was committed to version control in four steps: the data and its checks; then the pre-registration, holding the specification, all the code and the development-period forecast comparison, with no portfolio return computed; then the single run; then the robustness runs. The one choice made on data was the volatility model for the portfolios' scenarios: GARCH was to be chosen over EWMA only if its FZ0 loss, averaged across the ten portfolios, was lower on the development period with a Diebold-Mariano $p \leq 0.10$. It was, by -0.0282 ($t = -3.05$, $p = 0.002$).

Four changes were made after the plan was written and before the pre-registration, none using a forecast or portfolio return from the data: the exchange rate (above), the exact ES estimator (a unit test of the optimiser failed against the old one), the second stage of P1 (found on simulated returns while testing the run script), and the definitions of a budget breach and of six crisis windows. The audit and the replication in Sections 7 and 8 were specified in a second pre-registration before they were computed.

4. Results: forecasts of tail loss

Table 4 reports the five models in the development period and the holdout. Historical simulation hits close to the right number of days: its hit rate passes Kupiec's test in 9 of the ten portfolios in the holdout. Its hits come in clusters, so it passes the conditional-coverage test in only 4. Because its window is fixed and unweighted, it reacts to a rise in volatility only as the new days accumulate, and Figure 2 shows the result for 60/40 in 2008 and 2020: its forecast barely moved while the filtered forecast widened within days, and its breaches arrived in runs. The normal model understates the size of losses beyond VaR, and its Z_2 test rejects in 9 of ten portfolios. The filtered models pass the conditional-coverage test in 8 (EWMA) and 9 (GARCH and EVT) of ten.

On the FZ0 score every model beats historical simulation, in the development period with $p < 0.001$ and in the holdout with Holm-adjusted p of 0.033 for GARCH and EVT (Figure 3). The Basel traffic light (Basel Committee on Banking Supervision, 1996) tells the same story at 99%: in 18 blocks of 250 sessions of 60/40, historical simulation is green in 15, yellow in 2 and red in 1, filtered GARCH green in 17 and yellow in 1, and the normal model yellow in 10.

The counts in Table 4 overstate the evidence for calibration, because nine of the ten portfolios share assets. The joint test, which pools the ten and allows for their moving together, rejects every model, including filtered GARCH ($p = 0.015$) and GARCH-EVT ($p = 0.039$), though far less strongly than historical simulation (Wald statistic 15.8 against 61.3). The filtered forecasts are much better than historical simulation, but not exactly right. Every model fails at least one test on the gilt fund; Section 7 returns to it.

5. Results: the budget portfolios

Table 5 reports every rule in the holdout and over the whole period, and Table 6 the tests of the hypotheses.

Table 4. Forecasts of next-day VaR and ES at 97.5% for the ten test portfolios

Model	Pooled hit rate (%)	Pass conditional coverage (of 10)	Z2 rejects	Joint DQ *p*	FZ0 minus HS	Holm *p*
Panel A. Development, July 2008 to December 2016						
Historical simulation	2.33	5	1	< 0.001		
Normal, EWMA	2.96	6	8	0.008	−0.165	< 0.001
Filtered, EWMA	2.65	9	0	< 0.001	−0.185	< 0.001
Filtered, GARCH	2.50	10	0	0.013	−0.213	< 0.001
GARCH-EVT	2.44	10	0	0.010	−0.217	< 0.001
Panel B. Holdout, January 2017 to September 2026						
Historical simulation	2.85	4	5	< 0.001		
Normal, EWMA	3.38	4	9	< 0.001	−0.137	0.045
Filtered, EWMA	2.73	8	1	< 0.001	−0.175	0.037
Filtered, GARCH	2.72	9	1	0.015	−0.196	0.033
GARCH-EVT	2.55	9	1	0.039	−0.197	0.033

Pooled hit rate is the share of portfolio-days on which the loss exceeded the forecast VaR (2.5% if right). The next two columns count the portfolios that pass Christoffersen's conditional-coverage test at 5% and whose Z_2 test rejects at 5%. Joint DQ *p* is the pooled dynamic-quantile test of Section 7, which treats the ten portfolios as one body of evidence. FZ0 minus HS is the difference in average FZ0 loss from historical simulation (negative is better), with its Diebold-Mariano *p*-value corrected for four comparisons by Holm's method.

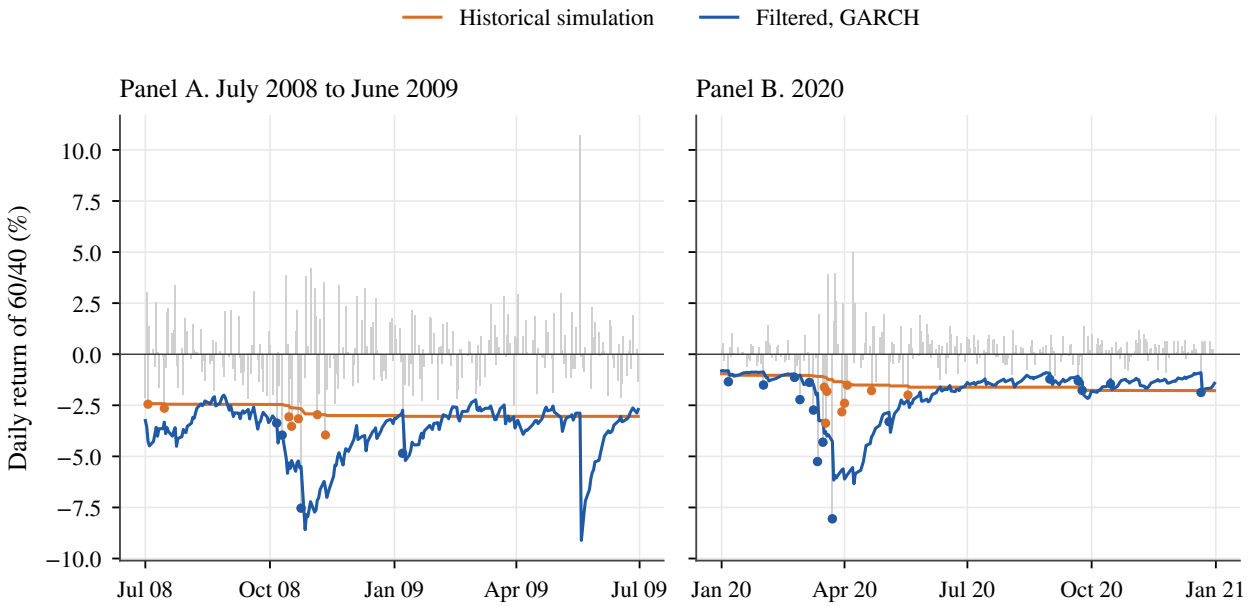


Figure 2. Next-day VaR forecasts at 97.5% for 60/40, made from returns before each day. Bars are daily returns; a dot marks a day whose loss exceeded the forecast of the same colour.

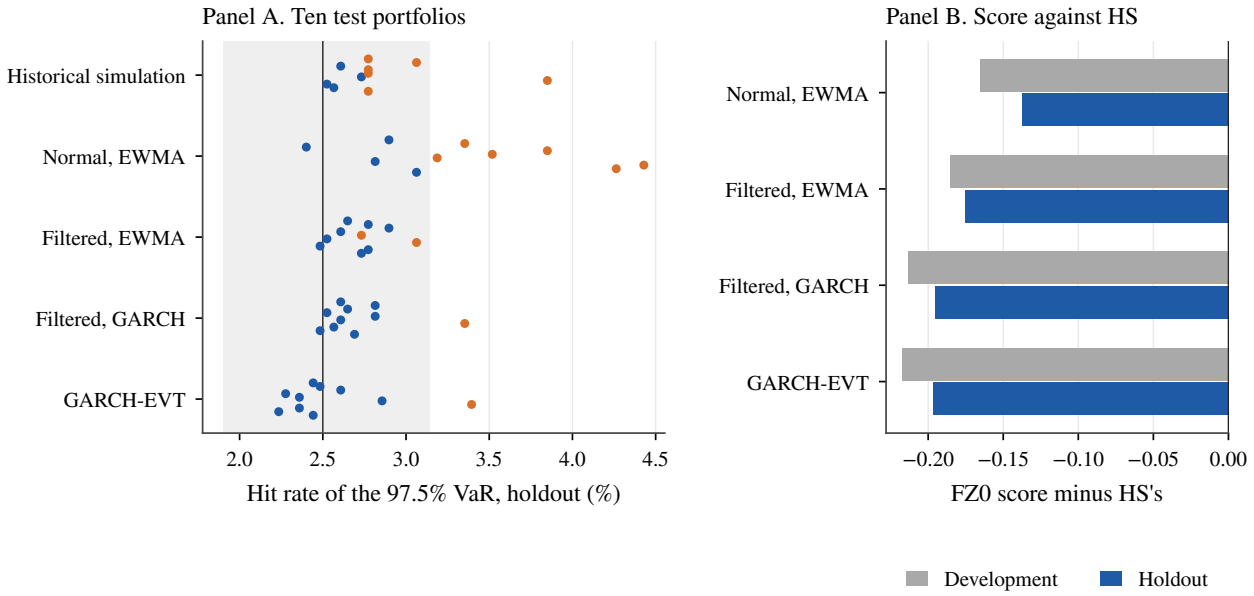


Figure 3. Panel A: hit rate of each model on each of the ten test portfolios in the holdout; orange marks a portfolio that fails the conditional-coverage test at 5%. The shaded band holds the hit rates that a correct forecast produces 95% of the time. Panel B: average FZ0 loss minus historical simulation's, in each period.

Table 5. The budget portfolios and their benchmarks, net of costs

Rule	Return (%)	Vol. (%)	Sharpe ratio	Max DD (%)	Realised ES (%)	Worst month (%)	Expo-sure	Turnover a year	Costs (%)
Panel A. Holdout, January 2017 to September 2026									
Budget optimiser, filtered (P1)	12.2	9.3	0.65	20.1	1.69	−7.0	1.00	3.49	0.91
Budget optimiser, historical (P2)	10.8	9.3	0.51	25.3	1.80	−14.8	0.99	0.31	0.12
Equal weight, scaled (P3)	12.8	7.4	0.86	13.0	1.38	−6.1	0.94	1.89	0.47
Risk parity, scaled (P4)	10.5	4.8	0.87	9.2	0.87	−4.1	1.00	0.20	0.06
Minimum CVaR, filtered (P5)	5.5	3.5	-0.15	7.0	0.64	−3.3	1.00	4.23	1.90
60/40, static	10.8	9.8	0.49	22.4	1.87	−12.9	1.00	0.07	0.01
Equal weight, static	14.5	8.6	0.93	20.7	1.66	−10.5	1.00	0.11	0.05
Equal weight, matched exposure	14.0	7.9	0.94	19.8	1.54	−9.5	0.92	0.07	0.03
Risk parity, matched exposure	10.8	5.0	0.89	10.5	0.91	−3.6	1.00	0.11	0.04
Panel B. Whole period, July 2008 to September 2026									
Budget optimiser, filtered (P1)	11.7	9.0	0.51	20.1	1.61	−7.0	1.00	2.89	0.79
Budget optimiser, historical (P2)	10.2	9.0	0.36	25.3	1.66	−14.8	0.99	0.33	0.12
Equal weight, scaled (P3)	12.3	7.5	0.67	13.0	1.33	−6.1	0.91	1.81	0.46
Risk parity, scaled (P4)	10.9	4.4	0.82	9.2	0.79	−4.1	1.00	0.25	0.06
Minimum CVaR, filtered (P5)	6.9	3.1	-0.04	7.3	0.60	−5.2	1.00	2.71	1.19
60/40, static	10.6	11.6	0.34	28.2	2.14	−15.2	1.00	0.07	0.01
Equal weight, static	13.5	9.4	0.67	24.4	1.79	−14.7	1.00	0.11	0.05
Equal weight, matched exposure	13.3	8.7	0.69	22.2	1.65	−12.5	0.92	0.14	0.05
Risk parity, matched exposure	11.0	4.5	0.83	10.5	0.81	−3.6	1.00	0.17	0.05

Returns are compound annual returns; the Sharpe ratio uses the liquid fund's daily return as the risk-free rate. Realised ES is the average of the worst 2.5% of daily returns over the period, as a loss; the budget is 1.5%. Exposure is the average share held in the eight risky assets. Turnover is one-way turnover a year, and costs are the return lost to them a year. The matched-exposure rows use the whole period's average exposure of P3 and P4 and are not tradable.

Table 6. Tests of the budget optimiser, P1

	Development	Holdout	Whole period
Own VaR forecast: hit rate (%)	2.62	2.52	2.57
Own forecast: conditional coverage p	0.87	0.93	0.96
Own ES forecast: Z2 (p)	−0.066 (0.23)	−0.064 (0.21)	−0.065 (0.15)
Realised ES (%)	1.51	1.69	1.61
95% interval (%)	1.36–1.66	1.42–2.02	1.44–1.80
p that ES exceeds the budget	0.43	0.11	0.12
Sharpe ratio, P1 / 60/40	0.35 / 0.22	0.65 / 0.49	0.51 / 0.34
Ledoit-Wolf p	0.66	0.51	0.40
Smallest detectable difference	0.84	0.71	0.56
Max drawdown, P1 / 60/40 (%)	10.8 / 28.2	20.1 / 22.4	20.1 / 28.2
Resamples with P1's drawdown smaller (%)	97	61	93
Budget breaches, P1 / P2	20 / 20	25 / 25	45 / 45
Breach after a breach, P1 / P2 (%)	10.0 / 5.0	8.0 / 24.0	8.9 / 15.6

P1's own forecast is the VaR and ES of the portfolio it held each day, from its conditional scenarios. The interval for realised ES and the drawdown comparison use a stationary bootstrap of daily returns (Politis and Romano, 1994). The Sharpe-ratio test is that of Ledoit and Wolf (2008). The smallest detectable difference is the annual Sharpe-ratio gap that a two-sided 5% test would detect with 80% probability. A budget breach is a day whose loss exceeded the budget itself, the same yardstick for P1 and P2.

5.1 Did the budget hold?

P1's own forecasts of its tail loss are calibrated in every period: its losses exceeded its forecast VaR on 2.52% of holdout days, and its conditional-coverage p is above 0.8 throughout. P2, the same optimiser on historical scenarios, has a hit rate of 3.06% and a conditional-coverage test result of $p < 0.001$ in the holdout. P1's realised ES was 1.51% in the development period and 1.69% in the holdout. The holdout excess is not significant: the bootstrap probability that it reflects a true ES above the budget has $p = 0.11$. The budget holds at each weekly decision, but between decisions prices drift and volatility moves, so the forecast ES of the portfolio actually held was above 1.5% on 30% of sessions, with a 95th percentile of 1.92%.

P1 met the budget by changing its mix, not by holding cash: it was at least 99% invested on 94% of sessions. When forecast risk rose it moved into gilt, gold and US Treasuries (Figure 4): 63% of the portfolio in September 2008, against 14% on average in 2019 and between 3% and 18% in January and February 2020. The COVID-19 crash took four weeks from a calm start, faster than weekly decisions and a GARCH forecast could respond, and P1 lost −17.1% in it.

5.2 Against 60/40

P1 returned 11.7% a year over the whole period with a Sharpe ratio of 0.51, against 10.6% and 0.34 for 60/40, and its maximum drawdown was 20.1% against 28.2% (Figure 5). In the holdout its Sharpe ratio was 0.65 against 0.49. None of these differences is significant (Ledoit-Wolf $p = 0.40$ over the whole period and 0.51 in the holdout), and the sample was too short for them to be: an 18-year sample can detect a Sharpe-ratio gap of 0.56 with 80% probability, against an observed gap of 0.18, and detecting the observed gap would take about 184 years of data. The comparison with 60/40 is not one this sample can settle.

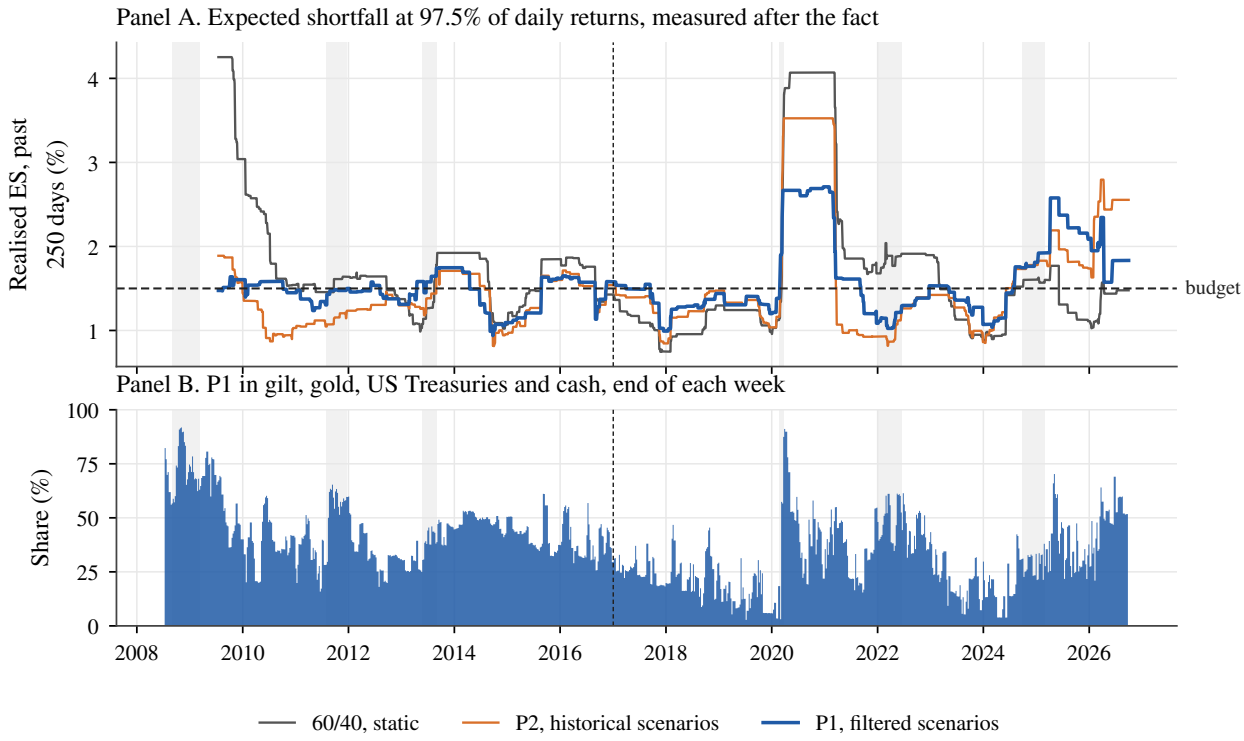


Figure 4. Panel A: expected shortfall at 97.5% of each portfolio's daily returns over the previous 250 sessions, measured after the fact, against the budget. Panel B: P1's weight in gilt, gold, US Treasuries and cash at the end of each week. Shading marks the six crisis windows; the dashed line marks the start of the holdout.

The crisis windows show where the difference came from (Figure 6). P1 lost less than 60/40 in five of the six windows, including −2.2% against −22.9% from September 2008 to March 2009, and more only in the rate rises of 2022, when bonds and equities fell together (−9.9% against −8.2%).

5.3 Filtered against historical scenarios

P1 did not breach its budget less often than P2: each lost more than 1.5% on 25 days in the holdout and 45 over the whole period. P1's breaches clustered less: the chance of a breach on the day after one was 8.0% for P1 and 24.0% for P2 in the holdout. P1 had the higher Sharpe ratio (0.65 against 0.51) and the smaller worst month (−7.0% against −14.8%), and lost less in 2020 (−17.1% against −24.6%), while P2 did better in 2008 (+4.0% against −2.2%). Filtering costs trading: P1 turned over its portfolio 2.9 times a year and paid 0.79% a year in costs, against 0.3 and 0.12% for P2.

5.4 What the benchmarks show

Static equal weight across the same eight assets did better than P1 on return and Sharpe ratio, 13.5% a year and 0.67 over the whole period. It did not keep to the budget: its realised ES was 1.79% and its worst month −14.7%. P1's objective rewards it for holding the most volatile assets the budget allows, 24% in the Smallcap 250 on average in the holdout, and in this sample that extra volatility was not rewarded.

Scaling a fixed mix by the forecast shortened its left tail at no cost to its Sharpe ratio. Equal weight scaled to the budget (P3) had a Sharpe ratio of 0.67 over the whole period against 0.69 for the same mix held at P3's average exposure ($p = 0.83$), a maximum drawdown of 13.0% against 22.2%, and a worst month of −6.1% against −12.5%. This is the pattern Harvey et al. (2018) report for volatility targeting; Section 8 finds that it

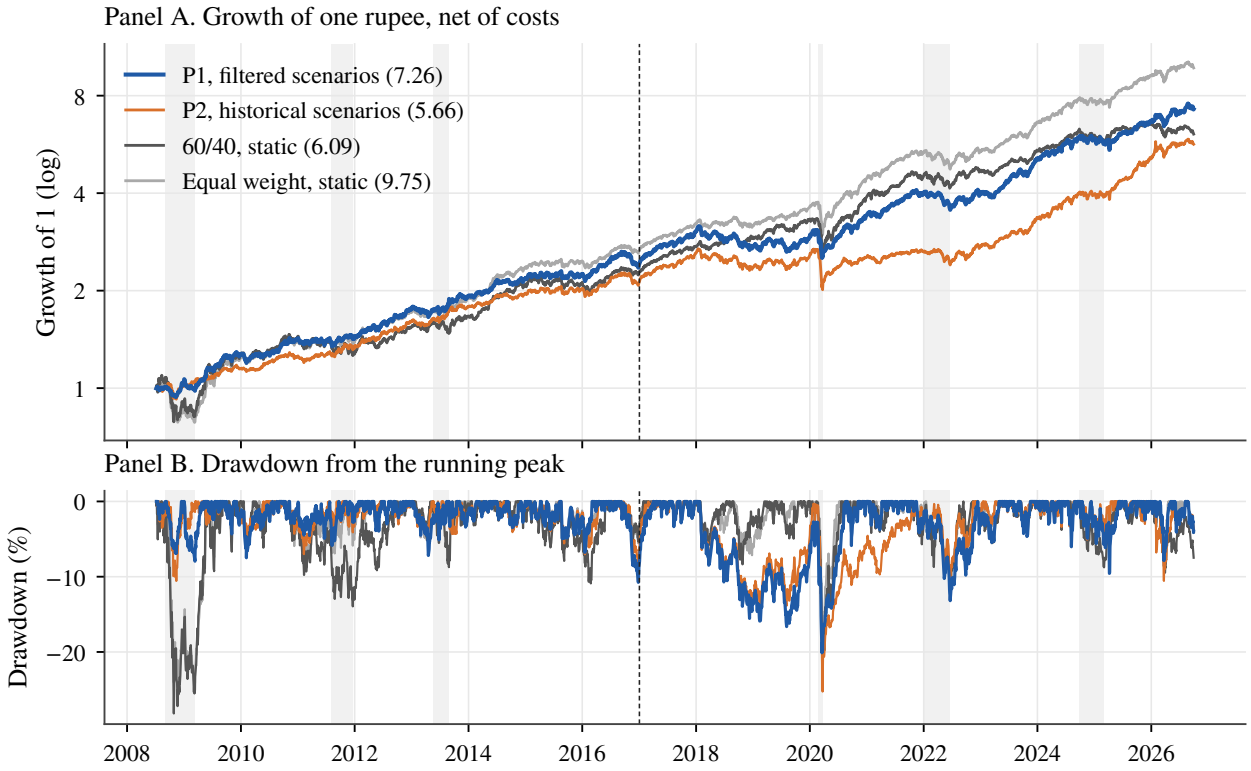


Figure 5. Growth of one rupee and drawdown of the budget optimisers and the static benchmarks, net of costs. The final values are in the legend. Shading marks the six crisis windows; the dashed line marks the start of the holdout.

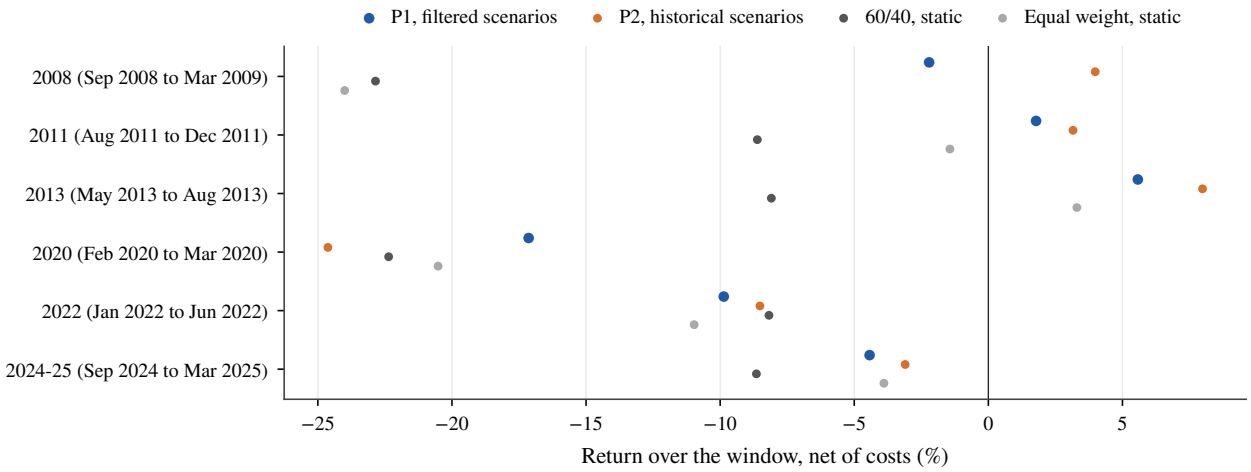


Figure 6. Return of each portfolio over the six crisis windows, net of costs. The windows were fixed before the single run. 2008, 2011 and 2013 lie in the development period and the rest in the holdout.

Table 7. Robustness of the budget optimiser

	Holdout				Whole period			
Variation	P1 SR	60/40 SR	LW p	P1 ES (%)	P1 SR	60/40 SR	P1 ES (%)	P1 DD (%)
Primary	0.65	0.49	0.51	1.69	0.51	0.34	1.61	20.1
ES at 95%	0.66	0.49	0.46	1.54	0.55	0.34	1.47	22.6
Budget 1.0%	0.62	0.49	0.65	1.20	0.57	0.34	1.16	12.9
Budget 2.0%	0.69	0.49	0.38	2.02	0.59	0.34	1.91	24.8
Monthly decisions	0.71	0.49	0.38	1.81	0.60	0.35	1.72	25.5
Costs doubled	0.55	0.49	0.81	1.70	0.43	0.34	1.61	21.5
1,000-day window	0.76	0.49	0.25	1.66	0.52	0.28	1.57	18.3
No 40% cap	0.56	0.49	0.77	1.68	0.43	0.34	1.61	21.4
EWMA filter	0.63	0.49	0.59	1.60	0.50	0.34	1.56	18.5
India only	0.79	0.49	0.20	1.64	0.52	0.34	1.57	16.9

Each row changes one setting of the specification. SR is the Sharpe ratio, LW p the Ledoit-Wolf p-value of the difference between P1 and 60/40, ES the realised ES at the variation's level, and DD the maximum drawdown. The 1,000-day window starts later, so its whole-period figures cover a shorter sample. ES at 95% keeps the 1.5% budget, which at 95% is a looser mandate.

does not carry over to the US before 2006. Risk parity sat inside the budget throughout, with a realised ES of 0.79%, so scaling it changed little.

Minimum CVaR on conditional scenarios (P5), which has no least-cost stage, turned over its portfolio 2.7 times a year and lost 1.2% a year to costs, 1.9% in the holdout, and earned less than cash. This is the fragility Lim et al. (2011) describe: as the scenarios change from week to week, the solution jumps.

6. Robustness

Each pre-registered variation in Table 7 changes one setting and reruns P1 and P2. None was used to choose anything. One variation does not apply: it changes the EWMA decay, and the filter chosen on the development period is GARCH.

P1's Sharpe ratio is above 60/40's in every variation and both periods, and never significantly; the smallest Ledoit-Wolf p is 0.19. Its realised ES follows the budget it is given, a little above it in the holdout at every level. Monthly decisions cut turnover from 2.9 to 1.3 a year but raised the holdout ES to 1.81%, so deciding less often holds the budget less well. The EWMA filter gives a similar result with turnover of 1.8 a year. With its trades held fixed, P1's costs could rise 2.6 times in the holdout and 3.0 times over the whole period before its Sharpe ratio fell to 60/40's. The breach-count result of the previous section does not hold up: over the whole period, P1 has more budget breaches than P2 in 5 of the nine variations.

7. An audit of the results

After the single run I specified four stricter tests, committed them before computing them, and recomputed the forecasts and P1 from the committed code; both reproduced the single run exactly. The tests can remove claims but not add them.

One joint test instead of a count. The dynamic quantile test of Engle and Manganelli (2004) regresses each hit minus 0.025 on a constant, four lags of the hit and the VaR forecast; if the forecasts are right, no coefficient differs from zero. I pool it across the ten portfolios, with standard errors that sum the scores

Table 8. Replication: forecasts on nine series, sessions before April 2006

Model	Pooled hit rate (%)	Pass conditional coverage (of 9)	Z2 rejects	Breach after a breach (%)	Joint DQ p	FZ0 vs HS, Holm p
Historical simulation	2.83	1	7	7.8–20.7	< 0.001	
Normal, EWMA	3.40	2	8	3.6–13.5	< 0.001	0.005
Filtered, EWMA	2.58	3	1	4.1–11.0	< 0.001	< 0.001
Filtered, GARCH	2.58	3	4	3.8–10.7	< 0.001	< 0.001
GARCH-EVT	2.52	3	0	4.0–10.0	< 0.001	< 0.001

The nine series are listed in the text. Breach after a breach is the range across the series of the share of hits that followed a hit on the previous day (2.5% if hits are independent). Joint DQ and FZ0 are pooled across the series trading each day.

across portfolios within each day before a Newey-West correction (Driscoll and Kraay, 1998), so portfolios that move together count as one piece of evidence on the days they do. It rejects every model in every period, as Table 4 shows; after Holm's correction (Holm, 1979) for five models, filtered GARCH and EVT are still rejected in the holdout ($p = 0.029$ and 0.039). The claim that the filtered forecasts are calibrated is therefore withdrawn. The claim that they beat historical simulation survives the same correction.

The gilt fund. As an exploratory check, specified after the audit's results were seen, I repeated the joint test without the gilt fund, which every model fails. In the holdout, filtered GARCH and EVT are then no longer rejected ($p = 0.065$ and 0.154), while the other three models still are. The gilt fund is not the only source of rejection, however: over the whole period GARCH is rejected without it as well ($p < 0.001$), and in the holdout the mid- and small-cap indices and US Treasuries reject GARCH on their own.

The budget. The bootstrap probability that P1's holdout ES exceeds the budget gives $p = 0.11$, and the Z_2 test of P1's own ES forecast, with p-values simulated from the scenarios of the portfolio it held each day, does not reject (Table 6).

Power. The smallest Sharpe-ratio difference against 60/40 that each period could detect with 80% probability is 0.84 in the development period, 0.71 in the holdout and 0.56 over the whole period, against observed differences of 0.13, 0.16 and 0.18.

8. Replication on data the model had not seen

The replication runs the method with every setting unchanged and makes no new choice. "Unseen" means every session before 3 April 2006, when Part II's panel begins; the years after that overlap in time with Part II and are reported separately.

8.1 Forecasts on nine long series

The nine series are the US equity market and the smallest 30% of US stocks by value (French, 2026), forecast from 1928; a 10-year US Treasury built from the 10-year yield and US 60/40, from 1964; the Nikkei 225, FTSE 100, DAX and Hang Seng; and the Nifty 50 total return index from 2001. The Treasury return is the price change of a 10-year par bond with yesterday's yield as its coupon, repriced at today's yield, plus the coupon accrued; its weekly correlation with IEF is 0.981 and its average calendar-year gap 0.96% outside 2003. The unseen period holds between 1,137 and 20,622 sessions per series, 86,440 in all.

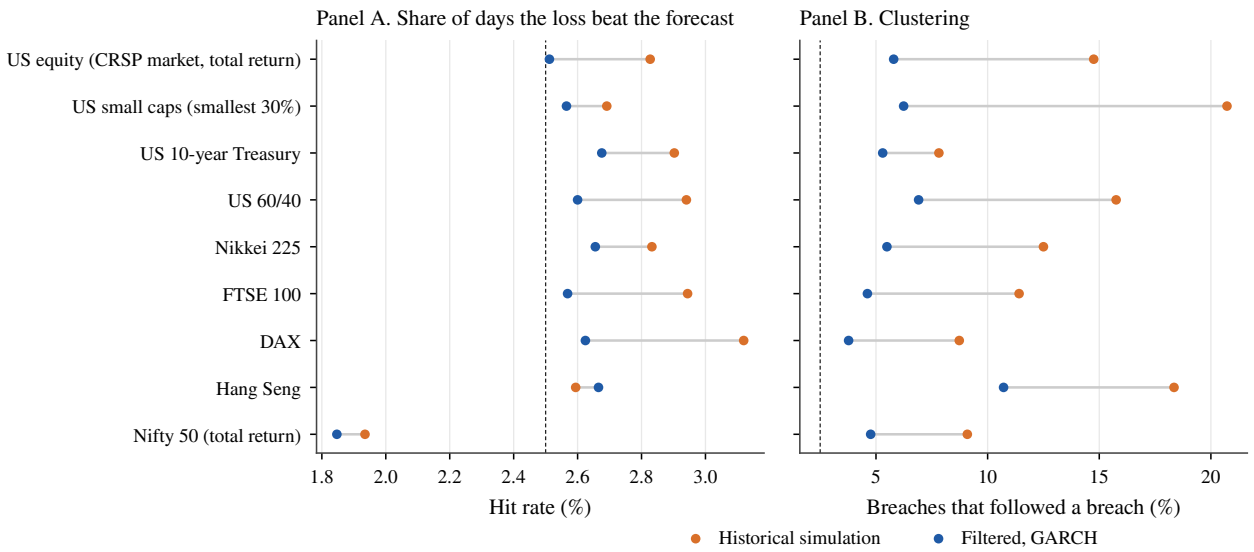


Figure 7. Replication of the forecasts, sessions before April 2006. Panel A: share of days on which the loss exceeded the 97.5% VaR forecast. Panel B: share of those days that followed a day with a breach. The dashed lines mark 2.5%, what a correct forecast gives in both panels.

Table 8 and Figure 7 report the results. The ranking replicates: every filtered model beats historical simulation on FZ0 with Holm-adjusted $p < 0.001$, and the share of breaches that follow a breach falls from between 7.8% and 20.7% for historical simulation to between 3.8% and 10.7% for GARCH. Calibration does not: with up to eighty years of daily data per series the tests have the power to reject every model on most series, and they do. GARCH's breaches still cluster most in the 1960s to the 1990s, when between 6% and 10% of them followed a breach. GARCH-EVT gets the size of the tail right: its Z_2 test rejects on none of the nine series.

8.2 The budget portfolio for a dollar investor

The dollar investor holds the US market, US small caps and the 10-year Treasury, with the one-month T-bill as cash, from 1962 to August 2026. Each side of a trade pays 10 basis points, and the benchmark is US 60/40 (60% market, 40% Treasury).

Table 9 and Figure 8 report the results. P1 held its budget: its realised ES was 1.40% before 2006 and 1.47% over the whole period, against 1.77% for 60/40, and it breached the budget less often than P2 in every period (117 against 146 days). It did not beat 60/40. Over 64 years its Sharpe ratio was 0.52 against 0.51 ($p = 0.83$), in a sample that could detect a gap of 0.18, and its deepest drawdown was 36.1% against 31.3%.

That drawdown came in 1973–74, when US stocks fell 48% from their peak at an annual volatility of 17%, about a third of the 55% of the 2008 crash. A budget on daily tail loss responds to how violently prices move, not to how far they fall, so P1 stayed 93% invested on average and lost −32.5% over the two years against −24.3% for 60/40. In crashes where volatility rose quickly it lost less: −3.9% against −10.5% from 2000 to 2002, −9.8% against −26.9% in 2008–09, −14.0% against −17.3% in 2020 and −16.0% against −22.0% in 2022. These US windows are descriptive and were not pre-registered.

P1's own forecast clusters in the US sample (conditional coverage $p < 0.001$ before 2006). Scaling equal weight by the forecast lowered its Sharpe ratio before 2006, 0.50 against 0.58 for the matched mix ($p = 0.048$), so the Indian result on scaling does not replicate.

Table 9. Replication: the budget portfolio for a dollar investor

	1964–2006 (unseen)	2006–2026 (overlap)	1964–2026 (all)
P1 return a year (%)	10.0	6.2	8.7
60/40 return a year (%)	9.7	8.5	9.3
Sharpe ratio, P1 / 60/40	0.52 / 0.43	0.53 / 0.65	0.52 / 0.51
Ledoit-Wolf p	0.26	0.29	0.83
Smallest detectable difference	0.22	0.32	0.18
P1 realised ES (%)	1.40	1.57	1.47
60/40 realised ES (%)	1.58	2.09	1.77
p that P1's ES exceeds the budget	0.98	0.18	0.79
Own VaR hit rate (%) and conditional coverage p	2.58 (< 0.001)	2.94 (0.07)	2.70 (< 0.001)
Max drawdown, P1 / 60/40 (%)	36.1 / 31.3	19.3 / 31.1	36.1 / 31.3
Budget breaches, P1 / P2	67 / 85	50 / 61	117 / 146
Breach after a breach, P1 / P2 (%)	9.0 / 11.8	8.0 / 4.9	8.5 / 8.9
Scaled EW / matched EW, Sharpe (p)	0.50 / 0.58 (0.048)	0.49 / 0.55 (0.562)	0.50 / 0.56 (0.247)

P1 and P2 are the rules of Section 3.4 with every setting unchanged, on three US assets with T-bills as cash. Scaled EW is equal weight scaled to the budget; matched EW is the same mix at its average exposure.

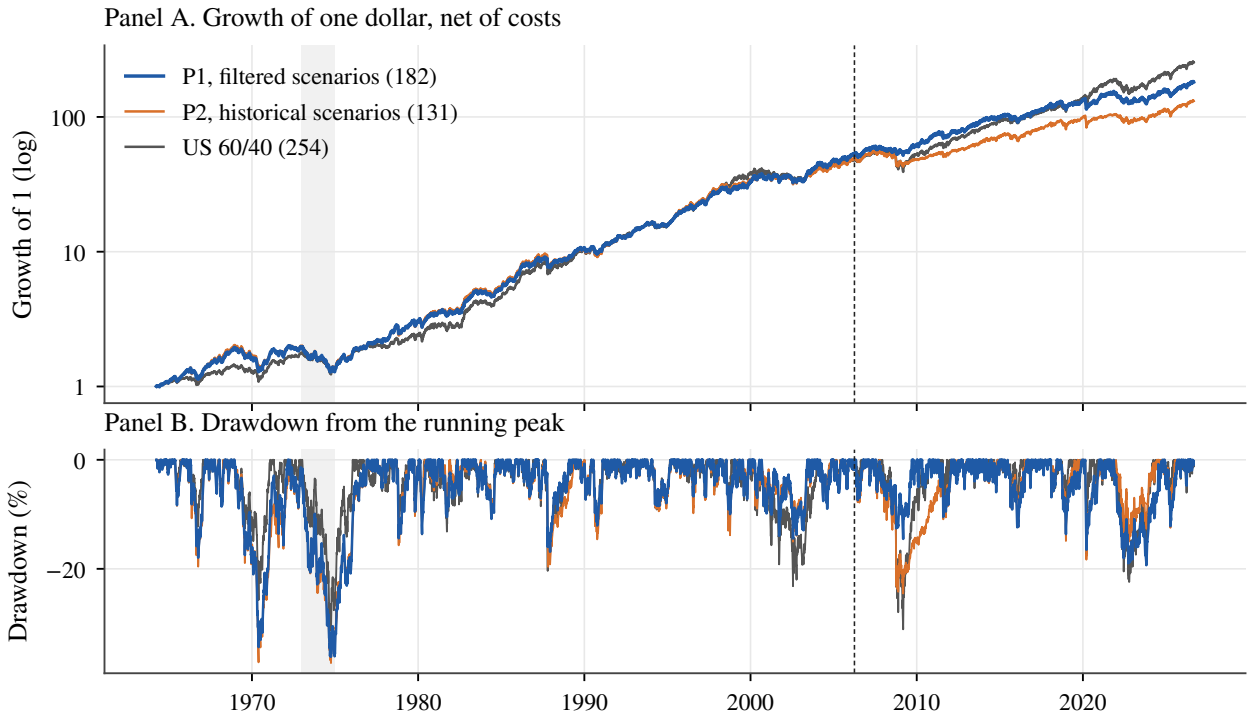


Figure 8. The dollar investor, 1964 to 2026: growth of one dollar and drawdown, net of costs. The dashed line marks April 2006; shading marks 1973–74.

9. Discussion

Three findings are consistent across all samples. First, forecasts that adjust historical returns for current volatility perform much better than forecasts that do not. Historical simulation gives equal weight to the most recent 500 days. As a result, after a calm period it tends to forecast calm conditions, and its breaches often occur in clusters. Filtering reduces this problem substantially for Indian assets since 2008 and for nine series dating back to 1928. It does not eliminate the problem: with long enough samples, every model is rejected. The evidence supports the conclusion that filtered models perform better, not that they are fully accurate.

Second, portfolios managed to a budget using these forecasts keep realised ES within about 0.2 percentage points of the budget in both markets. They also provide protection in the types of crashes they are designed to address: those in which volatility rises before most of the losses occur. Examples are 2008, 2000–02, 2020 and 2022 in the US, and 2008 and 2020 in India.

Third, maintaining the budget did not improve returns per unit of risk. In India, the estimated advantage over 60/40 was positive but not detectable over 18 years. Static equal weighting performed better, although it exceeded the budget. Over 64 years, the budget portfolio matched 60/40 for a dollar investor. A budget does not protect against a gradual decline with moderate volatility, such as 1973–74. In that case, the portfolio objective, which favours volatile assets, increased losses. A tail-loss budget is therefore a forecast-based risk limit, not a source of return.

The comparison with Part I is instructive. Part I asked which rule performed best, but the evidence did not identify a clear winner. Part II instead tested whether a rule delivers what it promises, using forecasts that can be evaluated. The results are mostly supportive.

10. Limitations

- *Significance.* Against 60/40 no difference is significant in the Indian sample, and the Indian sample could not have shown one of the size observed. The US sample could, and found none.
- *Calibration.* No model passes the joint calibration test on long samples. The budget is a forecast, not a guarantee: a crash from a calm start, as in 2020, gets through a weekly decision, and a slow decline does not trigger it.
- *Prior knowledge.* The literature already favoured filtered forecasts (Kuester et al., 2006) and volatility scaling (Harvey et al., 2018), the market history of 2008–2026 is common knowledge, and Part I's 2017–2026 results had been seen when the design was written.
- *Costs and taxes.* Costs are approximations, foreign trades in particular. Capital gains tax is ignored; P1's turnover of about three times a year would realise short-term gains, and remittance limits and the cap on Indian funds that invest abroad would make weekly trading in foreign assets impractical for many investors.
- *Data.* The funds used exist today, gold before 2007 is international gold in rupees, foreign prices are one day stale, which understates their correlation with Indian assets, and the US Treasury series of the replication is built from yields.
- *Independence.* The ten test portfolios share assets and the nine replication series fall together in crises; the joint tests allow for this within a day, but the series are not separate experiments.

11. Conclusion

A portfolio can be held to a tail-loss budget, in the sense that a forecast which rescales history by current volatility tells it how much risk to take, and its realised tail loss that stays close to the budget. That held for a rupee investor from 2008 to 2026 and for a dollar investor from 1964 to 2026, on data the method had not seen. What the budget did not do is improve return per unit of risk: it matched a 60/40 portfolio over 64 years and could not be distinguished from one over 18. It protected against crashes in which volatility rose quickly and not against a slow decline. The useful claim is the narrower one: tail risk can be forecast well enough to be managed, and historical simulation, the method of Part I, is not good enough to manage it.

Use of AI tools

The research question, the idea of the risk-budget design and of the asset universe and the interpretation of the results are my own. I used Claude Code, throughout this project, to learn more about computing, machine learning, and quantitative finance. In Part I it acted mainly as a tutor and wrote parts of the code at my direction. In Parts II and III, Claude almost acted as a peer; it shared ideas, criticised mine and helped me find a solution to a complex problem I could not solve on my own. Artificial intelligence has redefined "learning by doing" to a whole new extent; I have learned more about computing and quantitative finance through the implementation of this project compared to what any theory or literature on this subject would have taught me. Technical terms and concepts in the code and in finance that I was not familiar with, I learned through the explanations Claude provided. I asked Claude to explain each idea I had not used before, such as filtered historical simulation and the FZ0 score, and checked the explanations against the data before relying on them. I reviewed everything it produced and take full responsibility for the content. The process of reviewing the code, reporting the results and documenting the project alongside Claude's explanations gave me a system where I could implement what I learned in a practical manner, which made this project more insightful.

Data and code availability

The project repository, https://github.com/arjunprashanth6129/cvar-portfolio-optimizer, holds the data, code and results. Part I is in `src/`, and Part II, its audit and its replication are in `src/tailrisk/`, with the downloaded series and their sources in `data/tailrisk/raw/` and `data/tailrisk/replication/raw/`. The results memo, `results/tailrisk/RESULTS.md`, lists the commits in the order they were made. Running `python paper/make_assets.py` regenerates every number, table and figure in this paper from the committed output.

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